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Genetic optimization algorithm :

[redacted] :

Step 1 : Start
Step 2 : Generate an initial population randomly.
Step 3 : Evaluate the fitness of population.

Step 4 : Repeat until condition is statisfied.

4.a) Select best individual as parent.
4.b) ~~cos~~ Crossver b/w parents to create new offspring.
4.c) Mutate offspring to make small changes.
4.d) Check the fitness of new offspring
4.e) Create a new population by replacing some old individual with new offspring.

Step 5 : Return the best individual.
Step 6 : END

8/10/25

9) Traffic signal optimization :

→ Traffic congestion at road intersection causes long waiting time.

→ Objective is to optimize traffic signal timings based on traffic ~~[redacted]~~ condition, avg waiting time.

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Step 1 :- Initialize a population of traffic signal timings

Step 2 :- Calculate fitness based of avg. waiting time of vehicle

Step 3 :- Repeat until condition is satisfied.

→ Chose the best solutions as parent.
→ crossover & check for new traffic signal timing.
→ modify some timings.
→ ~~Select the best~~ Check the fitness of new population.
→ select the best solutions to form next gen.

Step 4 :- select the solution with minimum total Avg waiting timings.

Step 5 :- Return the optimized traffic signal timings.

Proceed & Right

Output :

Optimized traffic Signal Timings

North : 31 sec
South : 27 sec
East : 39 sec
West : 29 sec

Total Green time Time : 120 sec.
Estimated waiting score : 4.809.